

CHEMISTRY — SEMESTER - I

2 lesson(s) included, 1 missing

Included:

- Theme 2 : Elements, Compounds and Mixtures
- Theme 3 : Matter

MISSING (no downloaded Question Bank found):

- Theme 1 : *Introduction to Chemistry

Fill in the Blanks

1. A pure substance made up of only one kind of atom is called an ____.
2. A compound is formed by the ____ combination of elements.
3. The components of a mixture can be separated by ____ methods.
4. Salt dissolved in water forms a ____.
5. In a mixture, substances retain their ____ properties.

True or False.If false correct the statement.

1. Components of a mixture can be separated physically.
2. Elements can be broken down into simpler substances.
3. Sugar solution is a mixture.
4. A mixture has a fixed composition.
5. Salt and sand together form a mixture .

Multiple Choice Questions (MCQs)

- 1.Which of the following is an element?
(a) Water (b) Oxygen (c) Air (d) Salt
- 2.Which of the following is a compound?
(a) Iron (b) Copper (c) Water (d) Oxygen
- 3.Which of the following is a mixture?
(a) Carbon dioxide (b) Water (c) Air (d) Salt
- 4.Which method can be used to separate sand from water?
(a) Filtration (b) Burning (c) Melting (d) Freezing
- 5.Brass is made of:
(a) Copper and zinc (b) Iron and carbon
(c) Hydrogen and oxygen (d) Copper and tin

Short Answer Questions

1. What is an element?Give two examples of elements.
2. What is a compound?Give two examples of compounds.
3. What is a mixture?Give two examples of mixtures.
4. Define alloy.
5. Air a mixture .Give reason.
6. How can salt be separated from salt water?Name the process and Define it.

7. Water is a compound. Give reason.
8. Classify the following as element, compound, or mixture:
Gold , Air , Sugar , Salt water , Carbon dioxide

Long Answer Questions

1. Write any three characteristics of elements , compounds.
2. Compare elements, compounds, and mixtures in a table.
3. Explain any three methods used to separate mixtures.

Assertion–Reason Questions

- a. Both A and R are true and R is not the correct explanation of A
- b. Both A and R are true and R is the correct explanation of A
- c. A is true and R is false
- d. A is false and R is true

1. Assertion: Water is a compound.

Reason: Hydrogen and oxygen are chemically combined in a fixed ratio.

2. Assertion: Air is a mixture.

Reason: Its components can be separated by physical methods.

3. Assertion: Iron is an element.

Reason: It contains only one type of atom.

Answer the questions

1. Rice contains small stones mixed with it. Which method will you use to separate the stones and why?
2. A farmer wants to separate grains from the stalks after harvesting. Which method should be used?
3. Tea leaves need to be separated from prepared tea. Which method is most suitable?
4. Sand is mixed with water. How can the sand be separated from the water?
5. A mixture of iron filings and sand is given. How can the iron filings be separated?
6. Salt has been dissolved in water. How can the salt be obtained back?
7. A housewife wants to separate husk from wheat grains. Which method should she use?
8. Muddy water is collected after rain. How can clean water be obtained from it?

Derive the chemical formula of the following

1. Sodium has valency 1 and chlorine has valency 1.
2. Magnesium has valency 2 and oxygen has valency 2.
3. Calcium combines with chlorine. Calcium has valency 2 and chlorine has valency 1.
4. Aluminium has valency 3 and oxygen has valency 2.
5. A compound contains one calcium atom and two chlorine atoms.

Question 1: Assertion and Reasoning questions.

- a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true and R is not the correct explanation of A.
c) A is true but R is false
d) A is false but R is true.
- i) **Assertion (A):** Marble is metamorphosed limestone.
Reason (R): Marble is used to make floor tiles.
- ii) **Assertion (A):** Enamels are organic oil paints.
Reason (R): They are used to paint household articles.
- iii) **Assertion (A):** Sushruta is one of the most famous practitioners of Ayurveda.
Reason (R): Ayurveda is the world's oldest medical system.
- iv) **Assertion (A):** Chemistry plays an important role in our lives in many ways.
Reason (R): It aims at improving life quality.
- v) **Assertion (A):** Sodium benzoate used for preserving coloured fruits and vegetables.
Reason (R): It is added as a preservative in commercial pickling.

Question 2: Choose the correct answer.

- i) Proteins such as _____ fibres give strength to our body.
a) Collagen b) Sugar c) Oil d) Water
- ii) _____ is used to measure the temperature of substances.
a) Dropper b) Funnel c) Test tube d) Thermometer
- iii) Creams contain _____.
a) Zinc sulphate b) Zinc oxide c) Zinc chloride d) Zinc nitrate
- iv) Turmeric is a good natural _____.
a) Antacid b) Antiseptic c) Paint d) Enamel
- v) Cyclo Trimethylene Tri Nitramine is popularly known as _____.
a) SDX b) PVC c) RDX d) AGO
- vi) _____ are used in textile industry.
a) Plastics b) Soaps c) Dyes d) Glass
- vii) _____ is a natural ingredient that has antibacterial properties.
a) Clove b) Black pepper c) Cardamom d) Mustard

Question 3: State true or false. If false, correct the statement.

- i) Antoine Lavoisier published "Methods of Chemical Nomenclature".
- ii) Louis Pasteur developed a process called pasteurization.
- iii) Niels Henrik David Bohr made contributions in understanding the structure of an atom.
- iv) Glycerine is a chemical used to remove stains.
- v) Fertilizers make the soil infertile.
- vi) Glass is a non-crystalline material.
- vii) The herb tulsi is a good antipyretic.

Question 4: Match the following.

- i) Stain - Paper and cardboard
- ii) Antihistamines - Jellies and jams.
- iii) Cosmetics - Semi-synthetic fibre
- iv) Cellulose - Crush solid substances
- v) Pectin powder - Synthetic chemicals
- vi) Terrycot - Allergic reactions
- vii) Mortar - Discolouration
- Heating liquids

Question 5: Fill in the blanks.

- i) _____ acts as reservoir of energy.
- ii) Meat is preserved by _____ .
- iii) Aspirin and Paracetamol are commonly used as _____ .
- iv) Polystyrene is an example of _____ plastics.
- v) The cleaning property of _____ is stronger than that of soaps.
- vi) _____ are used as artificial implants of the body parts.
- vii) Mint leaves act as an _____ .

Question 6: Observe the picture and answer the following questions.

i)



- Name the major component of the items shown here.
- Write the full form of LPG and CNG
- List out the uses of LPG and CNG.

ii)



(A)



(B)



- Name the major component shown in the pictures (A) and (B). Give any two differences.
- Why is the component in (B) is more popular than the component in (A).
- What are the defects of the component in picture (B).

iii)



(A)



(B)



(C)

- Name the major component of the items shown here. Define the component.
- Name the chemicals present in the pictures A, B and C.
- How are synthetic chemicals harmful to us?

Question 7: Case study questions.

- A textile factory uses dyes to colour cotton fabrics.
 - What is a dye?
 - Why do dyes stick to fabrics?

- c) Give one example of a natural dye.
- ii) Riya learns that her stomach produces hydrochloric acid to help in digestion.
 - a) Where is hydrochloric acid found in the body?
 - b) How does it help in killing germs?
 - c) Name the areas it is used.
- iii) A builder uses limestone for making cement and marble for flooring.
 - a) Why is limestone used in making cement?
 - b) What is the main chemical in limestone and marble?
 - c) Give one use of limestone.
 - d) Is marble a natural or man made material?
- iv) A juice packet mentions preservatives on the label.
 - a) Why are preservatives added to packaged juice?
 - b) Why should we not consume too many preserved foods?
 - c) How can you identify if food contains preservatives?

Question 8: Answer the following.

- i) How chemistry plays an important role in defence? Give two examples.
- ii) What are drugs? List their types based on their usage.
- iii) Differentiate Plaster of Paris and cement.
- iv) What are thermosetting plastics? Give two examples.
- v) Differentiate soaps and detergents.
- vi) How fertilizers make the soil fertile?
- vii) Mention the uses of silicone.
- viii) Define a) Food processing b) Food preservation
- ix) List out the traditional and modern methods of food preservation.
- x) Why do Iron Pillar remains smooth and shining over years?
- xi) Name the chemicals and their use in preserving the food.
- xii) State the principle of food preservation.